HealthOps Analytics

DVA Capstone 2 • Section B, Group 11

Healthcare Resource Allocation & Patient Outcome Analysis

3 Hospitals • 5 Diagnoses • 4 Regions • 24,000 Patients

3 Interactive Tableau Dashboards (Executive • Risk • Cost)

Python · Jupyter · Tableau · GitHub

OVERVIEW

Agenda

01 Problem Statement & Objectives

02 Dataset & Tools

03 ETL Pipeline & Data Cleaning

04 Exploratory Data Analysis

05 Statistical Analysis

06 Key Findings

07 Tableau Dashboards

08 Business Recommendations

09 Limitations & Future Work

10 Team Contributions

PROBLEM STATEMENT

Why Does This Analysis Matter?

Inefficient healthcare resource allocation drives high costs, treatment delays, and poor patient outcomes.

Cost Variation

Treatment costs vary drastically across hospitals and diagnoses — up to 86% difference between lowest and highest diagnosis groups.

Treatment Backlog

12.56% of patients are under active treatment — Flu cases show the highest backlog at 14.44%, signalling staffing issues.

Outcome Disparity

Mortality rates and recovery rates differ by diagnosis and age group, suggesting gaps in care quality and triage protocols.

Data Silos

Patient data is spread across three hospital sources with inconsistent formats, missing values, and encoding issues.

DATASET & TOOLS

Dataset Overview & Technology Stack

24,000	3	5	4	25	0
Total Patients	Hospitals	Diagnoses	Regions	Source Columns	Missing (Post-Clean)

Python

ETL, EDA, Statistical Analysis, KPI computation using Pandas, SciPy, NumPy

Tableau

Interactive dashboards with hospital, diagnosis, region, and demographic filters

Jupyter Notebook

5 structured notebooks covering extraction → cleaning → EDA → stats → final load

GitHub

Version control with structured repo: data/, notebooks/, scripts/, docs/, tableau/

DATA PIPELINE

ETL Pipeline & Data Cleaning

5 notebooks | hospital_1.csv, hospital_2.csv, hospital_3.csv → 24,000-row clean dataset

01

Extraction

Ingested hospital_1, 2, 3 datasets → schema validation & initial profiling

02

Cleaning

Handled missing values, standardized formats, removed invalid cost rows, deduplicated patients

03

EDA

Generated KPI tables, correlation matrix, and key insight datasets

04

Statistical Analysis

Kruskal-Wallis, Chi-square, Spearman tests with effect size evaluation

05

Final Load Prep

Created hospital_tableau_ready.csv (25 columns) with analytical flags

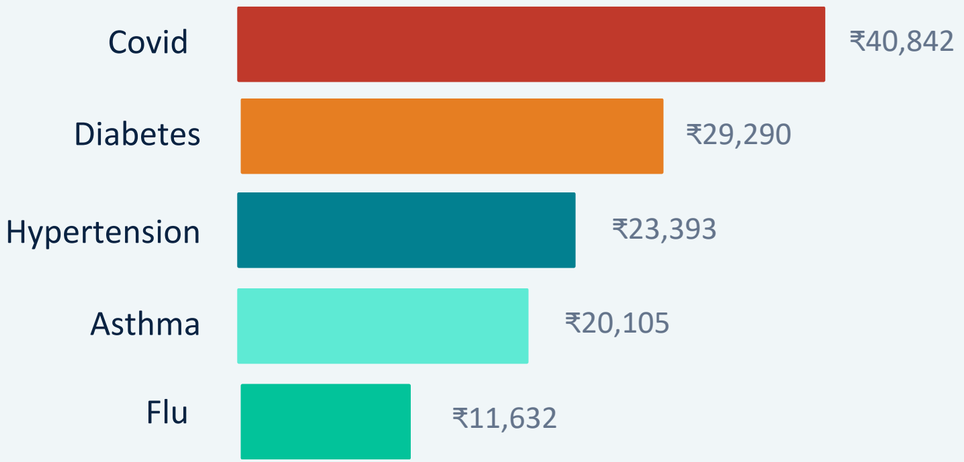
EXPLORATORY DATA ANALYSIS

EDA Highlights — Cost & Outcome Patterns

Average Treatment Cost by Hospital

Hospital	Type	Avg Cost (₹)	Recovery %
Sunrise	Private	28,504	82.56%
Carepoint	Private	24,119	82.56%
Medilife	Government	21,926	82.56%

Avg Treatment Cost by Diagnosis



Overall Patient Outcome Mix

82.6%
Recovered

12.6%
Under Treatment

4.9%
Deceased

STATISTICAL ANALYSIS

Statistical Validation of Key Patterns

Kruskal-Wallis · Chi-Square · Spearman Correlation · Mann-Whitney Pairwise Tests

Test	Variables	Statistic	p-value	Effect Size	Verdict
Kruskal-Wallis	Cost ~ Diagnosis	18,356	<0.001	0.73	SIGNIFICANT
Kruskal-Wallis	Cost ~ Hospital	1,048	<0.001	0.053	SIGNIFICANT
Kruskal-Wallis	Cost ~ Age Group	505	<0.001	0.022	SIGNIFICANT
Chi-Square	Outcome ~ Diagnosis	723.6	<0.001	0.123	SIGNIFICANT
Chi-Square	Outcome ~ Age Group	718.2	<0.001	0.122	SIGNIFICANT
Kruskal-Wallis	Cost ~ Insurance	—	0.278	—	NOT SIGNIFICANT
Spearman	Cost ~ Doc Availability	0.170	<0.001	0.170	WEAK POSITIVE
Spearman	Cost ~ Cleanliness Score	0.158	<0.001	0.158	WEAK POSITIVE

KEY FINDINGS

5 Evidence-Backed Findings

01 Diagnosis is the #1 Cost Driver

Cost varies significantly by diagnosis (Kruskal-Wallis, $p < 0.001$, effect size = 0.73). Covid treatment (₹40.8K) is ~86% higher than Flu (₹11.6K), driving most cost variation.

02 Covid is Highest-Cost & Highest-Risk

Covid shows the highest average cost (₹40.8K) and mortality (~11.9%). It is the only diagnosis with mortality exceeding 10%, requiring priority resource allocation.

03 Private Hospitals Have Higher Costs

Sunrise (₹28.5K) costs ~30% more than Medilife (₹21.9K). Cost differences across hospitals are statistically significant ($p < 0.001$), indicating pricing or case-mix variation.

04 Seniors Face Higher Mortality Risk

Outcome is significantly associated with Age Group (Chi-square $p<0.001$, Cramer's V 0.12). Senior patients show materially higher mortality than younger cohorts.

05 Flu Creates Maximum Treatment Backlog

Flu has the highest under-treatment rate (~14.4%) despite being lowest-cost, indicating operational bottlenecks in handling high-volume cases.

TABLEAU DASHBOARD

Dashboard Design & Visualisation Plan

hospital_tableau_ready.csv · 25 columns · 24,000 rows · 7 summary CSVs

Page 1	**Executive Overview**	Scorecard KPIs, Volume trend, Outcome donut, Region distribution, Diagnosis mix, Hospital type split, Demographics snapshot
Page 2	**Patient Outcomes & Risk**	Mortality KPIs (Senior, Covid, Low Income), Outcome by diagnosis (stacked), Outcome by age group, Mortality by economic status, Insurance vs outcomes
Page 3	**Cost & Financial Analysis**	Cost KPIs (Covid, Flu, Deceased), Treatment cost by diagnosis & hospital, Cost tier distribution, Financial distress by economic status, Recovery rate vs cost tier

Global Filters: Hospital · Diagnosis · Region · Age Group · Economic Status · Insurance

RECOMMENDATIONS

Data-Driven Business Recommendations

R1 Prioritize Covid Capacity & Protocols

Highest cost (₹40,842 avg) and highest mortality (11.94%). Standardise triage protocols and reserve capacity for Covid patients.

Evidence: KW stat 18,356 | p<0.001 | effect 0.73

R2 Audit Sunrise Cost Structure

Sunrise costs 30% more than Medilife. Investigate whether this reflects case mix differences, private pricing, or process inefficiency.

Evidence: KW stat 1,048 | p<0.001 | effect 0.053

R3 Reduce Flu Treatment Backlog

Flu has the highest under-treatment rate (14.44%) despite being the lowest-cost diagnosis. Allocate staffing and triage capacity.

Evidence: EDA: under-treatment rate 14.44% for Flu

R4 Monitor Senior Patient Risk

Senior patients show statistically higher mortality. Implement age-stratified escalation protocols and regular mortality review.

Evidence: Chi-square p<0.001 | Cramer's V 0.122

R5 Insurance is Not a Cost Driver

Insurance showed no statistical significance for cost (p=0.28) or outcome (p=0.55). Do not base resource decisions on insurance status.

Evidence: Chi-square p=0.55 | KW p=0.28 — NOT significant

R6 Operational Metrics as Monitoring Signals

Doctor availability and cleanliness score are statistically related to cost but effect sizes are small (rho ~0.16). Use as monitoring KPIs, not causal levers.

Evidence: Spearman rho 0.170 and 0.158 | p<0.001

Limitations & Scope for Future Work

Current Limitations

- Dataset shows identical outcome rates across hospitals — limits hospital-level outcome comparison.
- No causal inference — statistical associations only; confounders are not controlled.
- Limited model validation due to environment constraints
- Three simulated hospitals may not reflect the full diversity of real-world healthcare systems.

Future Work

→ Add time-series patient data to identify seasonal cost and volume patterns.

→ Introduce predictive modelling for patient mortality risk and cost estimation.

→ Expand to national-level dataset with 10+ hospitals across more states.

→ Integrate real-time Tableau extracts using scheduled pipeline refreshes.

TEAM DISTRIBUTION

Team Roles & Contributions — Group 11

Role	Primary Responsibility	Owner Name
Project Lead	Manages timeline, coordination, and ensures timely submission.	Manan Bansal
Data Lead	Sources data and maintains raw datasets and documentation.	Arjun Singh
ETL Lead	Handles data extraction, cleaning, and pipeline creation.	Shrihari K N
Analysis Lead	Performs EDA, statistical analysis, and KPI generation.	Udit Jain
Visualization Lead	Builds dashboards, prepares datasets, and finalizes charts.	Manan Bansal
Strategy Lead	Develops insights, recommendations, and narrative.	Harshita Joshi
PPT & Quality Lead	Prepares presentation, report, and ensures quality.	Pragya Kashyap

Thank You

Section B · Group 11 · HealthOps Analytics

24,000	5	13+	3
Patients Analysed	Notebooks Created	Statistical Tests	Dashboard Pages

Dashboard: tableu_dashboard_G_11

GitHub: github.com/Pinfinity07/SectionB_G11_HealthOps_Analytics